

Network Security

Lecture 3a

Wireless

Objectives

- Explain wireless technology
- Describe wireless networking standards
- Describe the process of authentication
- Describe wardriving
- Describe wireless hacking and tools used by hackers and security professionals

Belongs in action movies

- 2021 - Researchers found a **way to take control of a Tesla** by using a drone! The hack exploited Tesla's **Wi-Fi system**, which wasn't properly secured



"I was driving 70 mph on the edge of downtown St. Louis when the exploit began to take hold." - 2015

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Understanding Wireless Technology

- For a wireless network to function, you must have the right hardware and software
- Wireless technology is part of our lives
 - Baby monitors
 - Keyless entry systems
 - Cellphones and smartphones
 - GPS
 - Remote controls
 - Garage door openers
 - Two-way radios
 - Bluetooth speakers

Components of a Wireless Network

- A wireless network has only three basic components
 - Access Point (AP)
 - Wireless network interface card (WNIC)
 - Ethernet cable

Access Point

- An access point (AP) is a transceiver that connects to an Ethernet cable
 - It bridges the wireless network with the wired network
 - Not all wireless networks connect to a wired network
 - Most companies have Wireless LANs (WLANs) that connect to their wired network topology



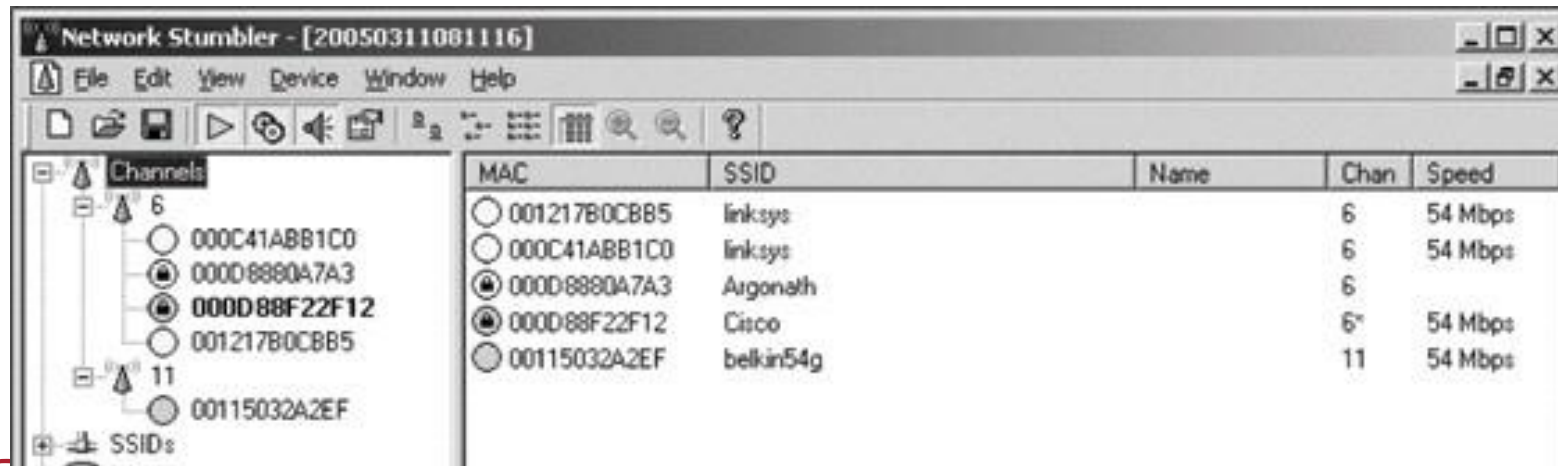
Netgear



Pineapple

Access Point

- The AP is where channels are configured
- An AP enables users to connect to a LAN using wireless technology
 - An AP is available only within a defined area



Service Set Identifiers (SSIDs)

- Name used to identify the wireless local area network (WLAN)
- The SSID is configured on the AP
 - Unique 1- to 32-character alphanumeric name
 - Name is case sensitive
- Wireless computers need to configure the SSID before connecting to a wireless network

Service Set Identifiers (SSIDs)

- SSID is transmitted with each packet
 - Identifies which network the packet belongs
- The AP usually broadcasts the SSID



Service Set Identifiers (SSIDs)

- Many vendors have SSIDs set to a default value that companies never change
- An AP can be configured to not broadcast its SSID until after authentication
 - Wireless hackers can attempt to guess the SSID

Table 11-1 Default SSIDs

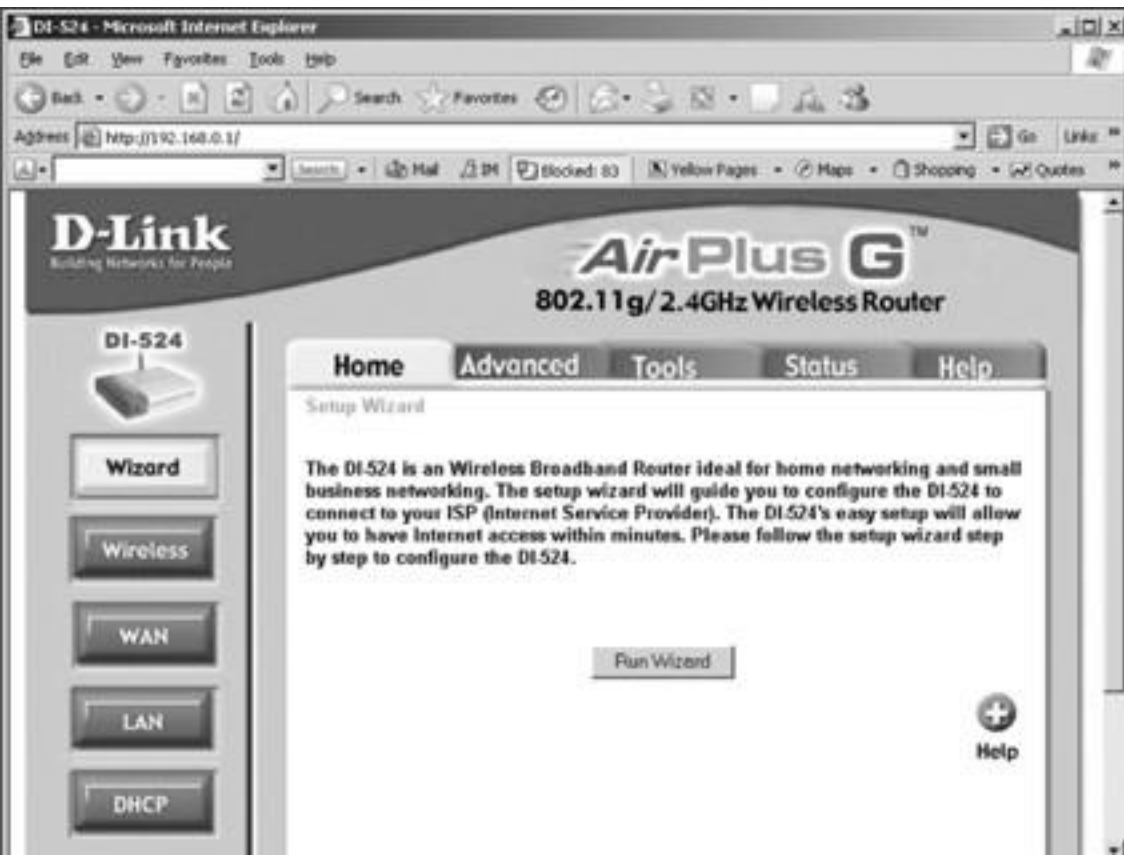
Vendor	Default SSIDs
3Com	3Com
Apple	Airport Network
Belkin (54G)	Belkin54g
Cisco	Tsunami
Compaq	Compaq
D-Link	WLAN, default
Dell	Wireless
Intel	Intel, 101, xlan, 195
Linksys	linksys, Wireless, linksys-g
Microsoft	MSNHOMe
Netgear	Wireless, NETGEAR
SMC	WLAN, BRIDGE, SMC
Symantec	101
US Robotics	WLAN, USR9106, USR808054

Configuring an Access Point

- Configuring an AP varies depending on the hardware
 - Most devices allow access through any Web browser
 - Enter IP address on your Web browser and provide your user logon name and password

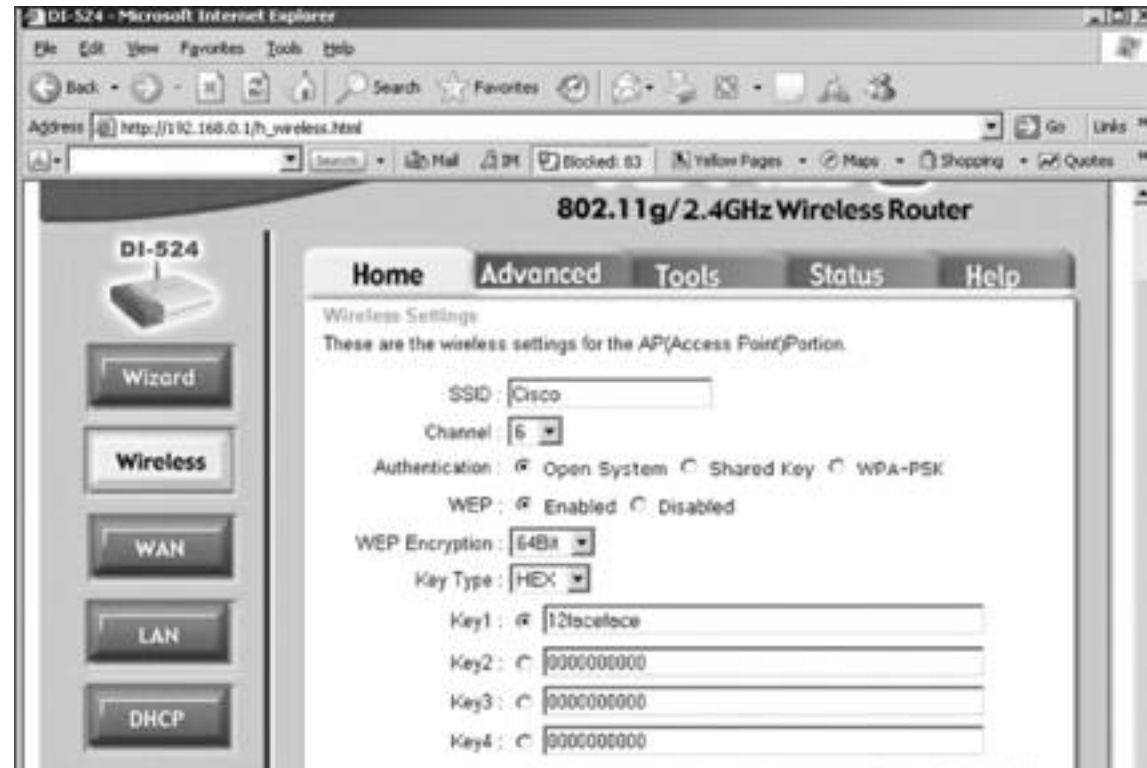
Wireless Router

- A wireless router includes an access point, a router, and a switch



Configuring an Access Point

- Wireless Configuration Options
 - SSID
 - Wired Equivalent Privacy (WEP) encryption
 - WPA (WiFi Protected Access) is better



Wireless NICs

- For wireless technology to work, each node or computer must have a wireless NIC
- NIC's main function
 - Converting the radio waves it receives into digital signals the computer understands

Wireless NICs

- There are many wireless NICs on the market
 - Choose yours depending on how you plan to use it
 - Some tools require certain specific brands of NICs
 - Cracking WEP requires NICs that can use monitor mode or perform packet injection

Understanding Wireless Network Standards

- A standard is a set of rules formulated by an organization
- Institute of Electrical and Electronics Engineers (IEEE)
 - Defines several standards for wireless networks

IEEE Standards

- Standards pass through these groups:
 - Working group (WG)
 - Sponsor Executive Committee (SEC)
 - Standards Review Committee (RevCom)
 - IEEE Standards Board
- IEEE Project 802
 - LAN and WAN standards

The 802.11 Standard

- The first wireless technology standard
- Defined wireless connectivity at 1 Mbps and 2 Mbps within a LAN
- Applied to layers 1 and 2 of the OSI model
- Wireless networks cannot detect collisions
 - Carrier sense multiple access/collision avoidance (CSMA/CA) is used instead of CSMA/CD

The Basic Architecture of 802.11

- 802.11 uses a basic service set (BSS) as its building block
 - Computers within a BSS can communicate with each other



The Basic Architecture of 802.11

- To connect two BSSs, 802.11 requires a distribution system (DS)

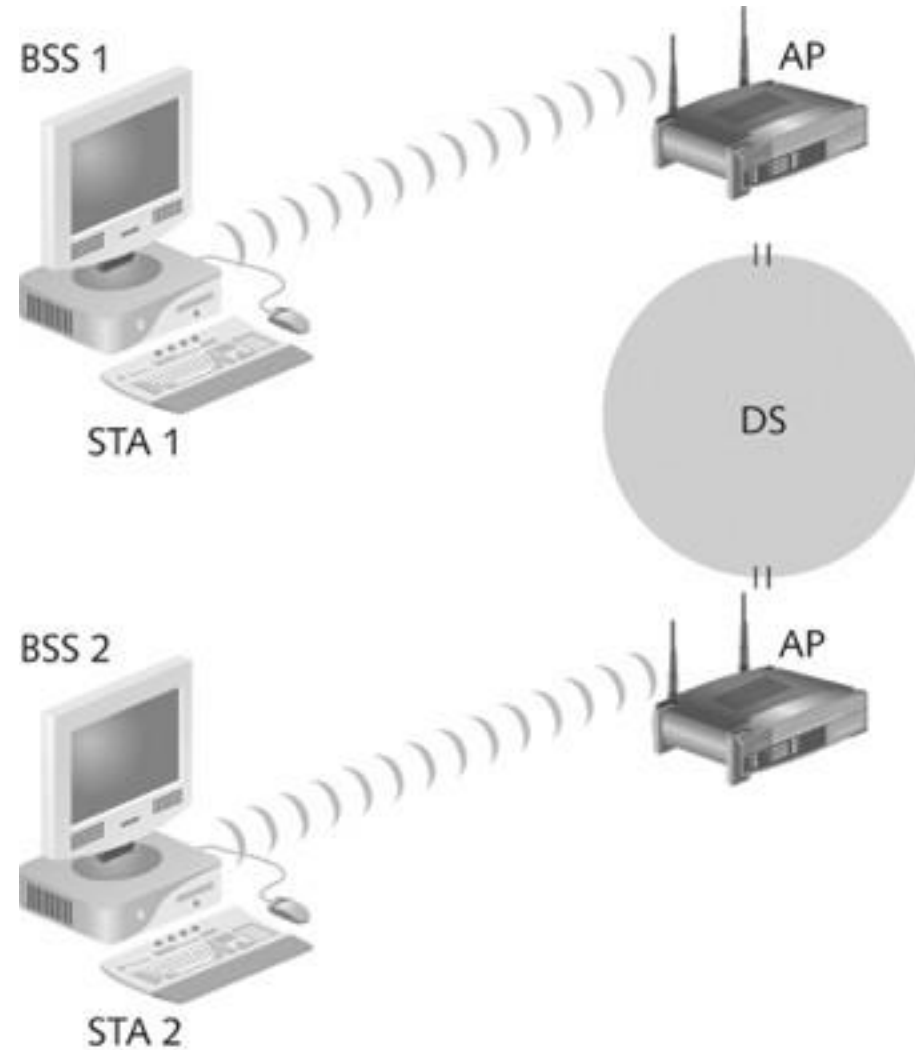
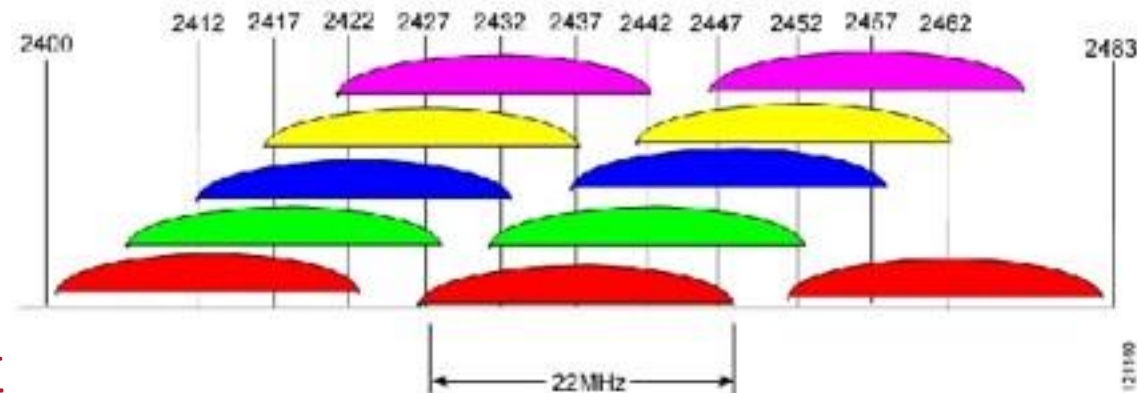


Figure 11-9 Connecting two wireless remote stations

Frequency Range

- In the United States, Wi-Fi uses frequencies near 2.4 GHz
 - (Except 802.11a at 5 GHz)
 - There are 11 channels, but they overlap, so only three are commonly used



802.11ac on MacBook Air

- Started with the 2013 model



802.11ad (WiGig)

- New physical layer
- 60 GHz, up to 7 Gbps
- Short-range (up to 30 feet)



TP-Link AD7200 Wireless Wi-Fi Tri-Band Gigabit Router (Talon AD7200)

by TP-Link



56 customer reviews

| 19 answered questions

Price: **\$349.99** ✓ Prime

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IEEE Additional 802.11 Projects (continued)

- HiperLAN2
 - European WLAN standard
 - It is not compatible with 802.11 standards

IEEE Additional 802.11 Projects (continued)

- 802.15
 - Addresses networking devices within one person's workspace
 - Called wireless personal area network (WPAN)
 - Bluetooth is one of six .15 standards
 - Image from ubergizmo.com



IEEE Additional 802.11 Projects (continued)

■ Bluetooth

- Defines a method for interconnecting portable devices without wires
- Maximum distance allowed is 10 meters
- It uses the 2.45 GHz frequency band
- Throughput of up to 2.1 Mbps for Bluetooth 2.0
 - Up to 24 Mbps for Bluetooth 3.0

IEEE Additional 802.11 Projects (continued)

- 802.16 (also called WiMAX)
 - Addresses the issue of wireless metropolitan area networks (MANs)
 - Defines the WirelessMAN Air Interface
 - Range of up to 30 miles
 - Throughput of up to 120 Mbps
- 802.20
 - Addresses wireless MANs for mobile users who are sitting in trains, subways, or cars traveling at speeds up to 150 miles per hour

Understanding Authentication

- Wireless technology brings new security risks to a network
- *Authentication*
 - Establishing that a user is authentic—authorized to use the network
 - If authentication fails, anyone in radio range can use your network

The 802.1X Standard

- Defines the process of authenticating and authorizing users on a WLAN
- Basic concepts
 - Point-to-Point Protocol (PPP)
 - Extensible Authentication Protocol (EAP)
 - Wired Equivalent Privacy (WEP)
 - Wi-Fi Protected Access (WPA)

Point-to-Point Protocol (PPP)

- Many ISPs use PPP to connect dial-up or DSL users
- PPP handles authentication with a user

name and password, sent with PAP or CHAP

- PAP (Password Authentication Protocol) sends passwords unencrypted
 - Vulnerable to trivial sniffing attacks

CHAP Vulnerability

- CHAP (Challenge-Handshake Authentication Protocol)
 - Server sends a Challenge with a random value
 - Client sends a Response, hashing the random value with the secret password
- This is still vulnerable to a sort of session hijacking attack
 - "Pass the hash"

Extensible Authentication Protocol (EAP)

- EAP is an enhancement to PPP
- Allows a company to select its authentication method
 - Certificates
 - Kerberos
 - Kerberos is used on LANs for authentication
 - Uses Tickets and Keys
 - Used by Windows 2000, XP, and 2003 Server by default
 - Not common on WLANS

LEAP

- Lightweight Extensible Authentication Protocol (LEAP)
 - A Cisco product
 - Vulnerable, but Cisco didn't care
 - Joshua Wright wrote the ASLEAP hacking tool to crack LEAP, and forced Cisco to develop a better protocol



More Secure EAP Methods

- Extensible Authentication Protocol-Transport Layer Security (EAP-TLS)
 - Secure but rarely used, because both client and server need certificates signed by a CA
- Protected EAP (PEAP) and Microsoft PEAP
 - Very secure, only requires server to have a certificate signed by a CA

802.1X components

Supplicant

- The user accessing a WLAN

Authenticator

- The AP

Authentication server

- Checks an account database to see if user's credentials are acceptable
- May use RADIUS (Remote Access Dial-In User Service)



Figure 11-11 A supplicant connecting to an AP and a RADIUS server

Wired Equivalent Privacy (WEP)

- Part of the 802.11b standard
- Encrypts data on a wireless network
- WEP has many vulnerabilities

Wi-Fi Protected Access (WPA)

- Specified in the 802.11i standard
- Replaces WEP
- WPA improves encryption by using Temporal Key Integrity Protocol (TKIP)

TKIP Enhancements

- Message Integrity Check (MIC)
 - Prevent attacker from injecting forged packets
- Extended Initialization Vector (IV) with sequencing rules
 - Prevent replays (attacker re-sending copied packets)

TKIP Enhancements

- Per-packet key mixing
 - MAC addresses are used to create a key
 - Each link uses a different key
- Rekeying mechanism
 - Provides fresh keys
 - Prevents attackers from reusing old keys

WPA Adds 802.1x

- WPA also adds an authentication mechanism implementing 802.1X and EAP
 - This was not available in WEP

WPA Versions

- WPA and WPA-2
 - WPA only implements part of 802.11i, using TKIP
 - Can run on older hardware
 - WPA2 implements the full IEEE 802.11i security standard, using CCMP
 - More secure

Pre-Shared Key v. 802.1x

- Both WPA and WPA-2 can run in either mode
- Pre-Shared Key uses a passphrase the user types into each device
 - Less secure because the user might choose a guessable passphrase, and all devices on the WLAN use the same passphrase
- 802.1x uses a server to manage keys
 - Each user has a different key
 - More secure

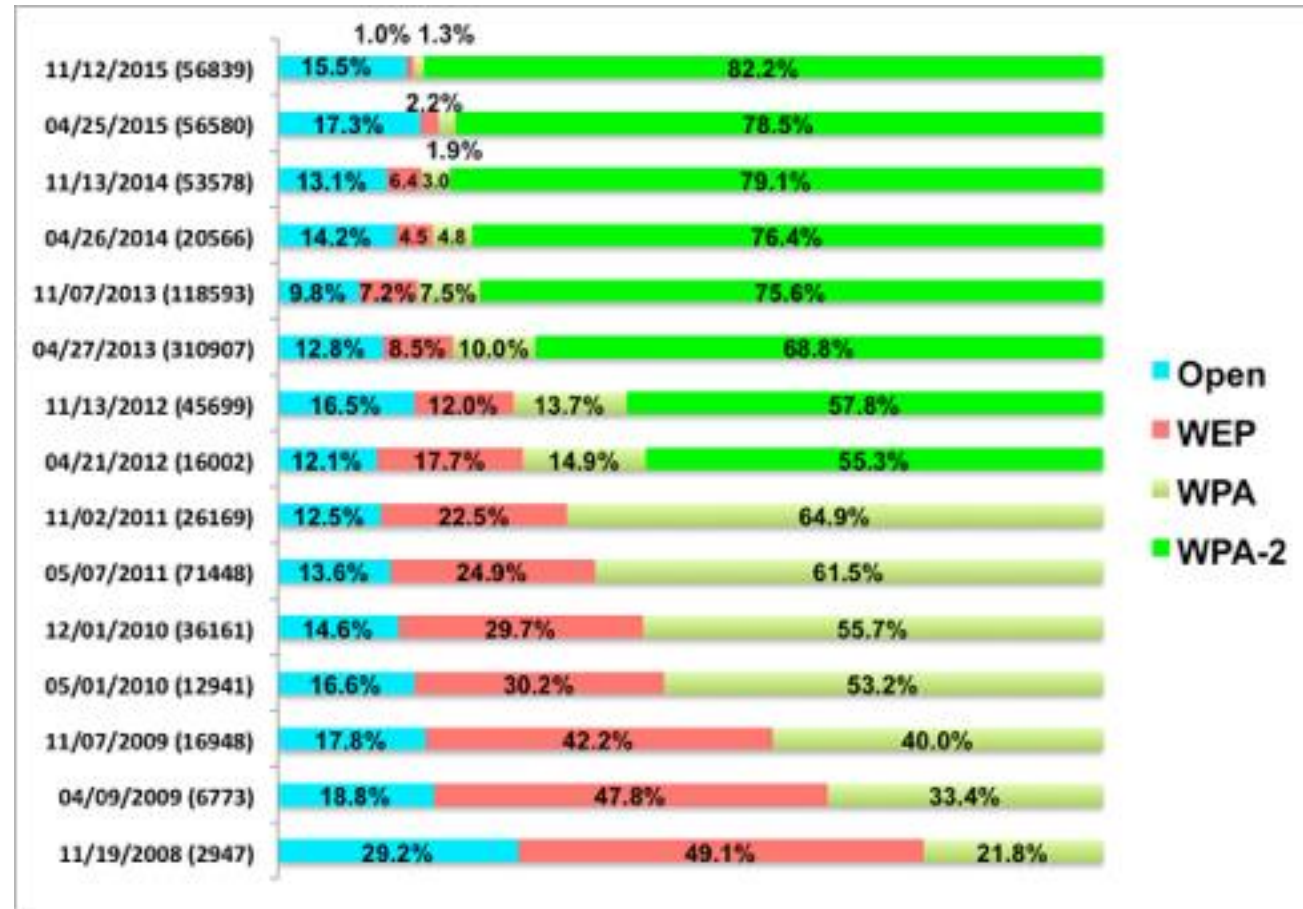
Hacking WPA

- In 2008 and 2009, some new WPA attacks were developed, for its weakest form (TKIP)
 - These attacks allow injection of spoofed packets for limited periods of time
 - Up to 18 minutes
 - They don't find the WPA key, but they are a warning that it's time to go to WPA-2

Understanding Wardriving

- Hackers use wardriving
 - Finding insecure access points
 - Using a laptop or computer
- Wardriving is not illegal
 - But using the resources of these networks is illegal
- Warflying
 - Variant where an airplane is used instead of a car

Wardriving results



How it works

- An attacker or security tester simply drives around with the following equipment
 - Laptop computer
 - Wireless NIC
 - An antenna
 - Software that scans the area for SSIDs
- Not all wireless NICs are compatible with scanning programs
- Antenna prices vary depending on the quality and the range they can cover

How It Works (continued)

- Scanning software can identify
 - The company's SSID
 - The type of security enabled
 - The signal strength
 - Indicating how close the AP is to the attacker

NetStumbler

- Shareware tool written for Windows that enables you to detect WLANs
 - Supports 802.11a, 802.11b, and 802.11g standards
- NetStumbler was primarily designed to
 - Verify your WLAN configuration
 - Detect other wireless networks
 - Detect unauthorized APs

NetStumbler

- NetStumbler is capable of interface with a GPS
 - Enabling a security tester or hacker to map out locations of all the WLANs the software detects

NetStumbler

- NetStumbler logs the following information
 - SSID
 - MAC address and Manufacturer of the AP
 - Channel
 - Signal Strength
 - Encryption (but not level of encryption)
- Can detect APs within a 350-foot radius
 - With a good antenna, they can locate APs a couple of miles away

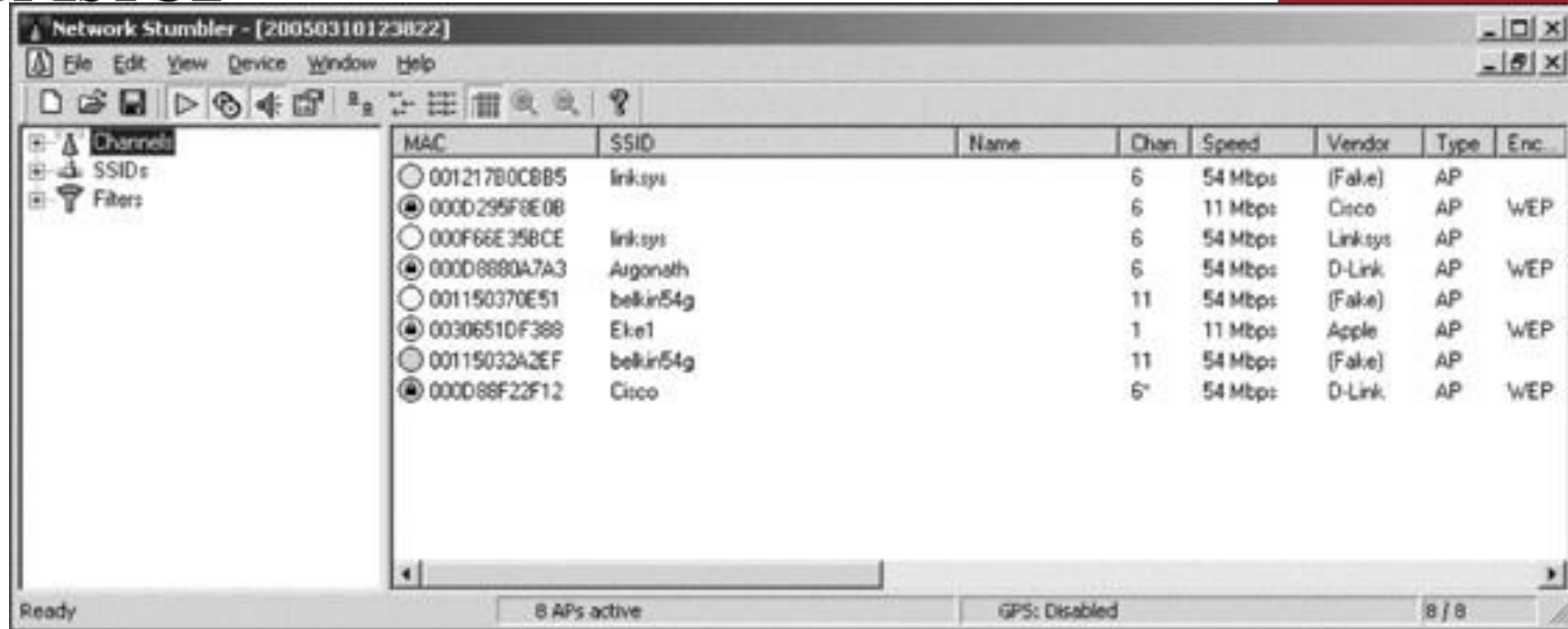


Figure 11-14 The Network Stumbler main window

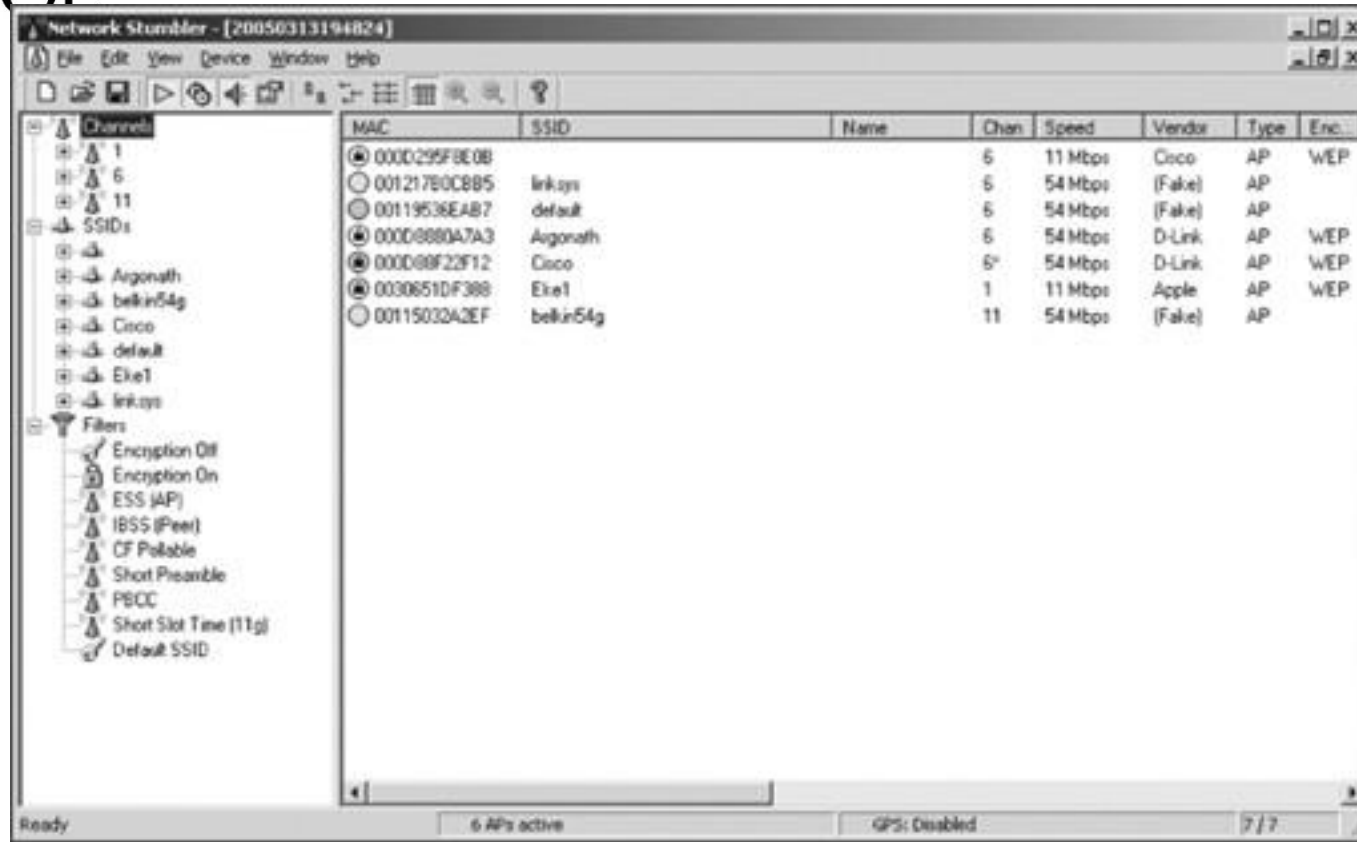


Figure 11-15 Viewing SSIDs, channels, and filters in Network Stumbler

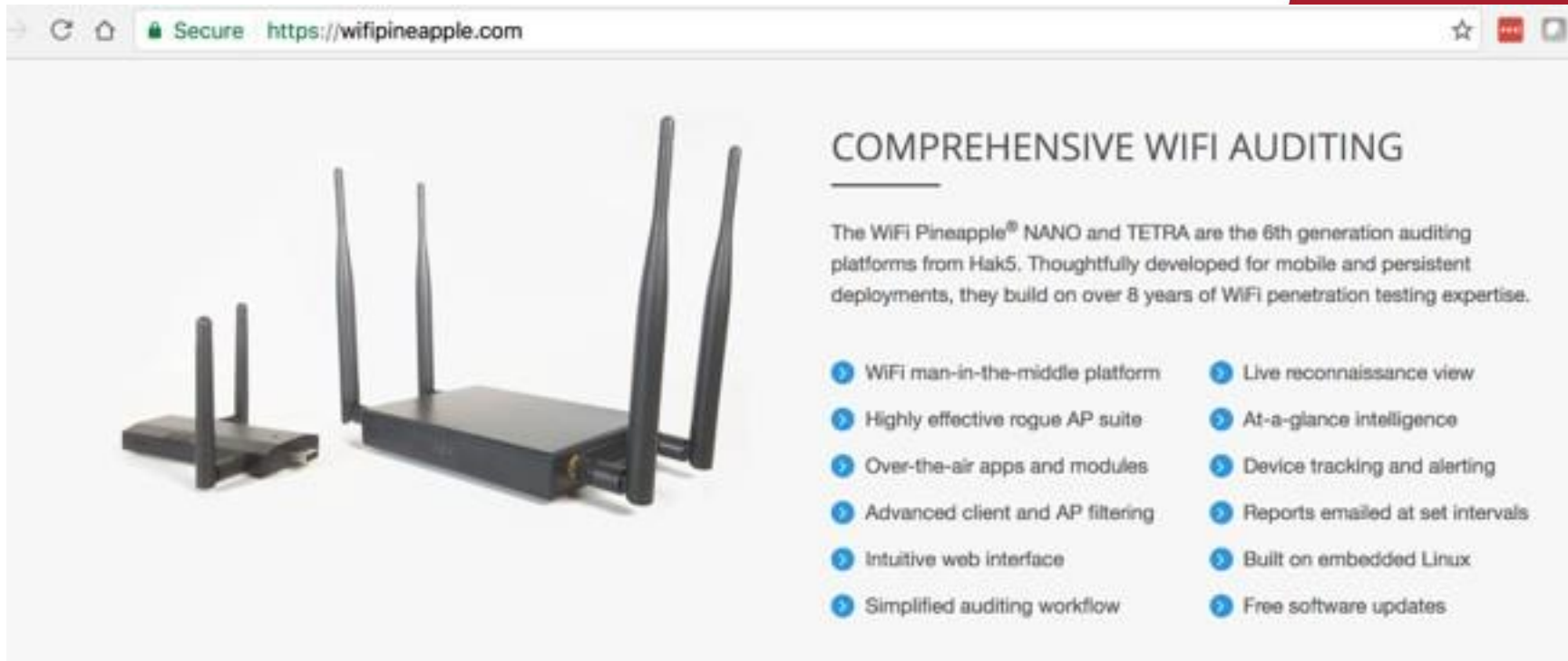
AirCrack NG

- Popular WEP-cracking tool
- Included in Kali

WEP-Cracking Project

- No longer worth the bother
- WEP is almost extinct

Wi-Fi Pineapple



The screenshot shows a web browser window with the address bar displaying "Secure https://wifipineapple.com". The main content area features a photograph of two Wi-Fi Pineapple devices: a smaller NANO model and a larger TETRA model, both with multiple antennas. To the right of the image, the heading "COMPREHENSIVE WIFI AUDITING" is followed by a paragraph describing the devices as 6th generation auditing platforms from Hak5. Below this, a list of features is presented in two columns, each item preceded by a blue arrow icon.

COMPREHENSIVE WIFI AUDITING

The WiFi Pineapple® NANO and TETRA are the 6th generation auditing platforms from Hak5. Thoughtfully developed for mobile and persistent deployments, they build on over 8 years of WiFi penetration testing expertise.

- WiFi man-in-the-middle platform
- Highly effective rogue AP suite
- Over-the-air apps and modules
- Advanced client and AP filtering
- Intuitive web interface
- Simplified auditing workflow
- Live reconnaissance view
- At-a-glance intelligence
- Device tracking and alerting
- Reports emailed at set intervals
- Built on embedded Linux
- Free software updates

Countermeasures for Wireless Attacks

- Anti-wardriving software makes it more difficult for attackers to discover your wireless LAN
 - Honeypots
 - Servers with fake data to snare intruders
 - Fakeap and Black Alchemy FakeAP
 - Software that makes fake Access Points

Countermeasures for Wireless Attacks

- Use special paint to stop radio from escaping your building
- Allow only predetermined MAC addresses and IP addresses to have access to the wireless LAN
- Use an authentication server instead of relying on a wireless device to authenticate users

Countermeasures for Wireless Attacks

- Use an EAP authentication protocol
- If you use WEP, use 104-bit encryption rather than 40-bit encryption
 - But just use WPA instead
- Assign static IP addresses to wireless clients instead of using DHCP
- Don't broadcast the SSID

